# Validating research questions on energy-efficient edge LLM inference

**Both proposed research questions are legitimate and grounded in real literature gaps, but they need sharpening to yield a focused, simulation-friendly term paper.** RQ1 (static vs. dynamic energy management) is well-motivated by at least five recent papers that independently demonstrate dynamic advantages, yet no single study systematically compares these strategies in the specific context of decentralized LLM inference with energy harvesting. RQ2 (comparing dynamic strategies) is the stronger and more novel question, but it currently lacks specificity about which dynamic strategies are being compared. Below is a full validation, literature map, gap analysis, and set of refined recommendations.

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## The reference paper provides a strong but narrow foundation

The Khoshsirat, Perin, and Rossi paper (GLOBECOM 2024) makes four contributions: empirical energy measurements on an NVIDIA Jetson AGX Orin, a semi-Markov model capturing battery state transitions across three variables (queue occupancy, discrete energy level, active/power-saving mode), three scheduling policies (Uniform, Long-term, Adaptive), and a dynamic power mode switching mechanism (ADS) that selects among **15W, 30W, and 60W** modes based on battery thresholds. (unibs) The dynamic mode achieved 47 completed jobs with 60% average battery over a fixed horizon, compared to 31 jobs/89% battery at fixed 15W and 58 jobs/16% battery at fixed 60W (unibs) — a clear Pareto improvement.

However, the paper has several exploitable limitations that create space for a term paper. The "LLM" workload is synthetic (100 encoder + 100 decoder layers), not a real model like LLaMA or Gemma. (unibs) Energy arrivals are i.i.d. uniform, ignoring diurnal solar patterns. (unibs) The network topology is minimal (3 groups × 3 devices). Communication overhead is entirely ignored. (unibs) The Adaptive scheduling heuristic lacks formal justification — it simply reduces input rates for devices in the lowest power mode by a manually chosen factor. (unibs) Most critically for your RQs, **the paper does not systematically compare its dynamic approach against a comprehensive set of static baselines**, nor does it compare across fundamentally different dynamic strategies (e.g., DVFS-based vs. scheduling-based vs. model-complexity-based adaptation).

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## RQ1 is valid but risks being already partially answered

The question of whether static or dynamic energy strategies are better for edge LLM inference with energy harvesting is supported by a genuine literature gap. At least five recent papers provide relevant but incomplete evidence. CLONE (Tian et al., USENIX ATC 2025) demonstrates

that **layer-wise RL-based DVFS achieves up to 7.36× energy savings** over static approaches for single-device LLM inference. (arXiv) (USENIX) DynamoLLM (Stojkovic et al., HPCA 2025) shows static cluster configurations are suboptimal under varying LLM workloads. (Jovans2) HarvNet (Jeon et al., MobiSys 2023) proves dynamic exit-point selection outperforms static policies on energy-harvesting devices running small DNNs. (ACM Digital Library) (arXiv) SparseDVFS (Zhang et al., 2026) demonstrates that operator-aware dynamic frequency scaling beats model-level static DVFS for edge inference. (arXiv) (arXiv)

The gap is that **none of these papers study the intersection of all three elements**: decentralized LLM inference, energy harvesting, and static-vs-dynamic comparison. The reference paper touches this intersection but only compares its own three policies rather than conducting a principled static-vs-dynamic analysis. RQ1 is therefore legitimate, but the risk is that the answer ("dynamic is better") is already predictable from existing evidence. The interesting questions are *by how much*, *under what conditions*, and *at what cost in complexity*.

**Verdict on RQ1:** Valid and well-motivated, but needs to be reframed around conditions and tradeoffs rather than a binary "which is better" question, since existing literature strongly suggests dynamic strategies dominate.

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## RQ2 is the stronger question but needs specificity

Comparing dynamic strategies is where the real novelty lies. The reference paper's Adaptive policy is only one type of dynamic strategy — a scheduling-level heuristic that redistributes load away from energy-poor devices. (unibs) But the literature reveals at least four distinct families of dynamic energy management:

- **Dynamic power mode switching** (the reference paper's approach): Adjusting device power mode based on battery level thresholds (unibs) (arXiv)
- **Dynamic scheduling/load balancing**: Redistributing inference requests across devices based on energy state (the Adaptive policy)
- **Dynamic frequency scaling (DVFS)**: Fine-grained voltage/frequency adjustment at the layer or operator level (USENIX) (arXiv) (CLONE, SparseDVFS)
- **Dynamic model complexity adaptation**: Adjusting quantization level, using early exits, or switching model variants based on energy budget (ACM Digital Library +2) (HarvNet, energy-adaptive checkpoint-free inference)

No existing paper compares these strategies against each other in a decentralized LLM inference setting. The reference paper's simulation framework could be extended to model at least the first three. For instance, you could compare: (a) the reference paper's battery-threshold-based power mode switching, (b) a greedy scheduling policy that always routes to the highest-energy device,

(c) an energy-proportional DVFS policy that continuously adjusts frequency rather than switching between discrete modes, and (d) a hybrid combining scheduling and power mode adaptation.

**Verdict on RQ2:** This is the more novel and interesting question, but it must specify *which* dynamic strategies are being compared. As currently stated, it is too vague to be actionable.

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## The literature reveals three major open problems worth targeting

**First, no work jointly optimizes scheduling and power mode adaptation for distributed edge LLM inference.** The reference paper treats these as separate mechanisms — power mode switching is a per-device rule, while scheduling operates at the network level. A joint optimization (or at least a systematic comparison of joint vs. decoupled strategies) would be novel and simulation-tractable.

**Second, the impact of realistic energy arrival patterns is unexplored.** Every paper in this space uses simplified energy models. The reference paper assumes i.i.d. uniform arrivals. unibs Real solar harvesting exhibits strong diurnal correlation, cloud-induced intermittency, and seasonal variation. arXiv Testing whether the reference paper's policies remain effective under correlated energy arrivals — and whether more adaptive strategies become necessary — is a clean, simulation-friendly research question.

**Third, sensitivity to system scale and heterogeneity is unstudied.** The reference paper uses 9 identical devices. unibs Real edge networks would feature heterogeneous hardware (different battery capacities, solar panel sizes, compute capabilities) and larger scale. Whether the Long-term and Adaptive policies degrade gracefully under heterogeneity is an open question.

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## 10 highly relevant related works for your term paper

**1. Khoshsirat, Perin, Rossi — "Decentralized LLM Inference over Edge Networks with Energy Harvesting"** (GLOBECOM 2024). Your reference paper. Semi-Markov model with three scheduling policies for battery-powered edge LLM inference. unibs arXiv Directly foundational to both RQs.

**2. Tian et al. — "CLONE: Customizing LLMs for Efficient Latency-Aware Inference at the Edge"** (USENIX ATC 2025). First paper applying layer-wise RL-based DVFS to LLM autoregressive inference. USENIX Achieves 11.92× speedup and 7.36× energy savings on Jetson Orin. arXiv Directly relevant to RQ1 (explicitly compares static vs. dynamic DVFS) and RQ2 (provides a dynamic DVFS strategy to compare against scheduling-based approaches).

**3. Stojkovic et al. — "DynamoLLM: Designing LLM Inference Clusters for Performance and Energy Efficiency"** (HPCA 2025). First energy-management framework for LLM inference clusters. Shows static configurations are suboptimal; proposes dynamic reconfiguration with multi-level controllers. Data-center focused but principles transfer to edge clusters. Supports RQ1's premise.

**4. Stojkovic et al. — "Towards Greener LLMs: Bringing Energy-Efficiency to the Forefront of LLM Inference"** (arXiv:2403.20306, 2024). Characterizes energy-performance knobs for LLM inference (GPU frequency, batch size, parallelism). Identifies that power management for LLM inference is largely unexplored. (arXiv) Motivational paper for both RQs.

**5. Jeon et al. — "HarvNet: Resource-Optimized Operation of Multi-Exit Deep Neural Networks on Energy Harvesting Devices"** (MobiSys 2023). RL-based scheduling that selects DNN exit points based on runtime energy harvesting status. (ACM Digital Library) Directly compares static vs. dynamic energy-aware inference policies. (arXiv) Relevant to RQ1 and provides a dynamic strategy template for RQ2 (energy-adaptive model complexity).

**6. Zhang et al. — "EdgeShard: Efficient LLM Inference via Collaborative Edge Computing"** (IEEE IoT Journal, 2024). Framework for partitioning LLMs into shards across edge devices and cloud. Uses dynamic programming for device selection and partition optimization. Relevant as a distributed LLM inference baseline, though it does not address energy harvesting.

**7. Borzunov, Ryabinin et al. — "Petals: Collaborative Inference and Fine-tuning of Large Models"** (ACL 2023 / NeurIPS 2023). Pioneering decentralized LLM inference system using BitTorrent-style protocol. Establishes the paradigm of distributing transformer layers across volunteer nodes. (arXiv) (unibs) Does not address energy — a gap your paper can reference.

**8. Zhao et al. — "LinguaLinked: A Distributed Large Language Model Inference System for Mobile Devices"** (ACL 2024). Decentralized LLM inference on mobile devices with optimized model assignment and runtime load balancing. (ACL Anthology) Tested on Pixel 7 Pro and CUBOT X30. (arXiv) Relevant as a practical distributed edge LLM system without energy considerations.

**9. Husom et al. — "Sustainable LLM Inference for Edge AI: Evaluating Quantized LLMs for Energy Efficiency"** (ACM TIoT, 2025). Benchmarks 28 quantized LLMs on Raspberry Pi 4 with hardware power measurement. (arXiv) (ACM Digital Library) Shows quantization reduces energy by up to **79%**. Provides essential energy baselines for edge LLM inference. Relevant as a static energy optimization approach for RQ1.

**10. Islam et al. — "Energy-Adaptive Checkpoint-Free Intermittent Inference for Low Power Energy Harvesting Systems"** (IEEE, 2025). Dynamic model complexity reduction when harvested energy is limited, plus checkpoint-free progress preservation during power failures. (arXiv +2) Directly addresses dynamic energy adaptation for EH devices, relevant to both RQs as a strategy that could be adapted for LLMs.

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### Recommended path: merge and refine the two RQs into one focused question

Pursuing both RQs separately in an 11-page two-column paper will spread you too thin. RQ1's answer is largely predictable (dynamic wins), and RQ2 is too vague without specifying strategies. Instead, I recommend **merging them into a single, sharper research question** that subsumes both:

> **Refined RQ: "How do different energy utilization strategies — static power mode selection, threshold-based dynamic power switching, and adaptive scheduling — compare in throughput, energy efficiency, and device availability for decentralized LLM inference on energy-harvesting edge networks, and under what conditions does the added complexity of dynamic approaches yield meaningful gains?"**

This refined RQ has several advantages. It naturally includes the static-vs-dynamic comparison (RQ1) while also comparing specific dynamic strategies against each other (RQ2). It is directly answerable via simulation by extending the reference paper's semi-Markov framework. It produces results with clear practical implications — system designers need to know when simple static policies suffice versus when dynamic adaptation is worth the overhead.

**Concrete simulation plan for this refined RQ:**

- **Baseline strategies**: (1) Fixed 15W mode with uniform scheduling, (2) Fixed 30W with uniform scheduling, (3) Fixed 60W with uniform scheduling — these are pure static baselines
- **Semi-static strategies**: (4) Long-term scheduling with fixed power mode, (5) Long-term scheduling with dynamic power mode switching
- **Dynamic strategies**: (6) Adaptive scheduling with dynamic power mode (the reference paper's best approach), unibs (7) A new energy-proportional strategy where scheduling probability is continuously weighted by current battery level rather than using the reference paper's discrete heuristic
- **Sensitivity analyses**: Vary energy arrival rates (from energy-scarce to energy-abundant), introduce correlated energy arrivals (simulating diurnal solar patterns), scale the network (from 9 to 27 devices), and introduce device heterogeneity

This gives you a clean **7-strategy comparison** across **4 experimental dimensions**, all within the reference paper's simulation framework with modest extensions. The paper structure would be: introduction (1 page), background and related work (2 pages), system model and strategies (2 pages), experimental setup (1 page), results and analysis (3.5 pages), conclusion (0.5 pages) — fitting comfortably in 11 two-column pages.

**Alternative RQ if you want something more exploratory:** *"How sensitive are energy-aware scheduling policies for decentralized edge LLM inference to the statistical properties of energy arrivals — specifically, does temporal correlation in energy harvesting (diurnal patterns) degrade the performance of policies optimized for i.i.d. arrivals, and can simple adaptive modifications recover performance?"* This is narrower but highly novel, since no existing work tests these policies under realistic energy profiles.

**Bottom line:** RQ2 is stronger than RQ1, but the best approach is to merge both into a single comparative study. The reference paper's simulation is directly extensible for this purpose, and the resulting paper would make a clear contribution by providing the first systematic comparison of static and dynamic energy strategies for this emerging application domain.